EXPERIMENT 9

Query with Joins – Equi Join, Inner Join, Outer Join

AIM:

To perform JOIN using Equi Join, Inner Join, Outer Join on the given relation.

DESCRIPTION:

JOIN

A MySQL join is a method of linking data from one or more table based on values of the common column between tables.

MySQL supports the following types of joins:

1. Cross join
2. Inner join
3. Left join
4. Right join

CROSS JOIN

The CROSS JOIN makes a Cartesian product of rows from multiple tables. Suppose, you join t1 and t2 tables using the CROSS JOIN, the result set will include the combinations of rows from the t1 table with the rows in the t2 table.

INNER JOIN

To join two tables, the INNER JOIN compares each row in the first table with each row in the second table to find pairs of rows that satisfy the join-predicate. Whenever the join-predicate is satisfied by matching non-NULL values, column values for each matched pair of rows of the two tables are included in the result set.

LEFT JOIN

Unlike an INNER JOIN, a LEFT JOIN returns all rows in the left table including rows that satisfy join-predicate and rows do not. For the rows that do not match the join-predicate, NULLs appear in the columns of the right table in the result set.

RIGHT JOIN

A RIGHT JOIN is similar to the LEFT JOIN except that the treatment of tables is reversed. With a RIGHT JOIN, every row from the right table (t2) will appear in the result set. For the rows in the right table that do not have the matching rows in the left table (t1), NULLs appear for columns in the left table (t1).

QUESTIONS

1. List the **patient names and their appointment details** by joining the Patient and Appointment tables.

2. Find the patients who **do not have any appointment**. Display the Patient ID and Patient Name.

3. List the **doctor names and appointment details** for doctors who have appointments.

OUTPUT:

1)

```sql
SELECT Patient.Patient_ID,
       Patient.Patient_Name,
       Appointment.App_ID,
       Appointment.Doctor_ID,
       Appointment.App_Date,
       Appointment.App_Time,
       Appointment.Room_No
FROM Patient
INNER JOIN Appointment
ON Patient.Patient_ID = Appointment.Patient_ID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Patient_ID	Patient_Name	App_ID	Doctor_ID	App_Date	App_Time	Room_No
P001	Arun Kumar	A001	D001	2026-08-10	10:00:00	R101
P002	Priya	A002	D002	2026-08-10	11:00:00	R102
P003	Rahul	A003	D003	2026-08-11	10:30:00	R103
P004	Sneha	A004	D004	2026-08-11	12:00:00	R104

2)

```sql
SELECT Patient.Patient_ID,
       Patient.Patient_Name
FROM Patient
LEFT JOIN Appointment
ON Patient.Patient_ID = Appointment.Patient_ID
WHERE Appointment.Patient_ID IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Patient_ID	Patient_Name

3)

```sql
SELECT Doctor.Doctor_ID,
       Doctor.Doctor_Name,
       Appointment.App_ID,
       Appointment.Patient_ID,
       Appointment.App_Date,
       Appointment.App_Time,
       Appointment.Room_No
FROM Doctor
INNER JOIN Appointment
ON Doctor.Doctor_ID = Appointment.Doctor_ID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Doctor_ID	Doctor_Name	App_ID	Patient_ID	App_Date	App_Time	Room_No
D001	Ravi	A001	P001	2026-08-10	10:00:00	R101
D002	Priya	A002	P002	2026-08-10	11:00:00	R102
D003	Kumar	A003	P003	2026-08-11	10:30:00	R103
D004	Anitha	A004	P004	2026-08-11	12:00:00	R104

RESULT:
The records from the tables are displayed using JOIN using EquiJoin, InnerJoin, OuterJoin.